

UMS Multi-Tag & Influx Repopulation Studio

Quick Operations & Setup Guide | Zero-Dependency Interactive Web UI

1. Set Credentials & Test → 2. Upload CSV & Preview → 3. Run Dry Run → 4. Live Execution & Influx Backfill

1. Quick Start & Access

- **Option A (One-Click):** Double-click `run-ui.bat`
- **Option B (Command Line):** Run `python ui_server.py`
- **Browser URL:** Open `http://127.0.0.1:8080`

2. CSV Input File Requirements

The system requires a CSV file with 3 standard columns:

Column Header	Description	Example Value
MeterName	Meter Display Name or GUID in UMS	EM-1121 MCHW-FWU-22-01104
MainTag	Root tag category / namespace	test-tag-version-1
taghierarchy	Full tag path under the main category	test-tag-version-1/test-tag-5

```
MeterName,MainTag,taghierarchy
EM-1121 MCHW-FWU-22-01104,test-tag-version-1,test-tag-version-1/test-tag-5
EM-1121 MCHW-FWU-22-01104,testing,testing/testtag-4
EM-1074 TX-E-01002,test-tag-version-1,test-tag-version-1/test-tag-6
```

Multi-Tagging: If the same meter appears on multiple rows with different tags, the engine automatically merges them into a single update payload.

3. Step-by-Step UI Workflow

Step 1: UMS Connection

- Enter **Base URL** (e.g. `https://api-ums.iviva.com`), **Account Key**, and **API Token**.
- Click **Test Connection** to verify API connectivity.

Step 2: CSV Upload & Preview

- Select or drag your CSV file into the upload zone.
- Click **Preview CSV** to verify grouped meter counts and tag associations.

Step 3: Repopulation Parameters

- **Trigger Influx Repopulate:** Checked by default (sends `isChange=1`).
- **Start Time:** ISO 8601 UTC timestamp for backfill window (e.g., `2026-05-31T18:30:00.000Z`).

- **Stop Time:** ISO 8601 UTC timestamp (defaults to current time).

Step 4: Dry Run vs. Live Execution

- **Start Dry Run:** Validates tags and matches CSV meters against UMS with *zero modifications*.
- **Start Live Execution:** Creates missing tags, merges existing meter tags, patches meters, and initiates InfluxDB series backfilling.

4. REST API Reference

Action	Method	Endpoint	Description
Get Tags	GET	/api/v1/tags	Retrieves existing tag hierarchy
Create Tag	POST	/api/v1/tags	Creates missing root or sub-tags
List Meters	GET	/api/v1/utilitymeters	Resolves meter names to meter IDs
Patch & Repopulate	PATCH	/api/v1/utilitymeters/{id}	Applies tags & triggers Influx repopulate (<code>isChange=1</code>)

Safety Guarantee: Existing meter tags (e.g. `energy/cooling` , `location` , `ghg`) are preserved and merged.